

Formulas:-

① $P(A) = \frac{n(A)}{n(S)}$

② $P(A') = 1 - P(A)$

③ $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Mutually Exclusive
↓

$P(A \cup B) = P(A) + P(B)$

④ If A & B are independent

$P(A \cap B) = P(A) P(B)$

⑤ $P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)}$

(conditional)

⑥ Bayes →

$P(A_i | B) = \frac{P(B | A_i) P(A_i)}{\sum P(B | A_j) P(A_j)}$

⑦ Odds in favour of A is a:b

$P(A) = \frac{a}{a+b}$

Odds in against of A is a:b

$P(A) = \frac{b}{a+b}$

⑧ ${}^n P_r = \frac{n!}{(n-r)!}$

⑨ ${}^n C_r = \frac{n!}{r!(n-r)!}$

(10) P.M.F:-

$$P(X=x)$$
$$\sum P(X=x) = 1$$

(12) C.D.F:-

$$F(x) = P(X \leq x)$$
$$F(x) = \int_{-\infty}^x f(t) dt$$

(14) 2nd Moment:-

$$E(X^2) = \sum x^2 P(X=x)$$
$$E(X^2) = \int x^2 f(x) dx$$

(16) $\sigma = \sqrt{\text{Var}(X)}$

(11) Continuous R.V:-

$$f(x) \geq 0$$
$$\int_{-\infty}^{\infty} f(x) dx = 1$$

(13) Mean:-

Discrete

$$E(X) = \sum x P(X=x)$$

$$E(X) = \int x f(x) dx$$

Continuous

(15) Variance

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

Binomial

(17) $P(X=x) = {}^nC_x p^x (1-p)^{n-x}$

Mean $= E(X) = np$

Variance $= \text{Var}(X) = npq$

(18) Poisson

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Mean $= E(X) = \text{Var}(X) = \lambda$

(19) Normal

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\text{Z-score} = \frac{x-\mu}{\sigma}$$

$$X \geq a \Rightarrow Z = \frac{a - \frac{1}{2} - \mu}{\sigma}$$

(Binomial)
(Continuity Correction)

(20) n th Moment about origin.

$$\mu_r' = E(x^r)$$

n th Central Moment:-

$$\mu_r = E[(x - \bar{x})^r]$$

$$\mu_2 = \text{var}(x)$$

(21) Correlation:-

Karl Pearson Correlation:-

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Regression:-

(22) Y on X

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$b_{yx} = r \frac{\sigma_y}{\sigma_x}$$

X on Y

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$b_{xy} = r \frac{\sigma_x}{\sigma_y}$$

$$r = \sqrt{b_{xy} b_{yx}}$$

(23) Curve fitting

St. Line

$y = a + bx$ / Exponential fit

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

Parabola:-

$$y = a + bx + cx^2$$

$$y = ab^x$$

$$\log y = \log a + x \log b$$

(24)

Sampling:-

Standard Error of Mean: $(SE) = \frac{\sigma}{\sqrt{n}}$

Z-test: $(Z) = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$

(25)

Grouped Data:-

$\bar{X} = \frac{\sum f x}{\sum f}$

Median = $l + \frac{\frac{N}{2} - c.f}{f} \times h$

Mode = $l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$

* Definitions:-

- ① Random Experiment ② Sample Space ③ Event,
- ④ Classical Probability, ⑤ Outcome, ⑥ Impossible Event
- ⑦ Certain Event ⑧ Complementary Event
- ⑨ Mutually Exclusive ⑩ Equally Likely Event.
- ⑪ Exhaustive Event ⑫ Axiomatic Probability.
- ⑬ Compound Probability ⑭ Dependent Events.
- ⑮ Bayes's Theorem.

Proofs / Conversions:-

- ① Addition, Multiplication Theorem. ② Bayes's Theorem
- ③ Variance ④ Mean, Variance of Binomial.
- ⑤ Poisson as limit of Binomial.
- ⑥ Mean of Poisson. ⑦ Area under Normal Curve.
- ⑧ $|r| \leq 1$. ⑨ Regression Coefficient ⑩ C.R.

(11) Poisson $\rightarrow$ Binomial.

(12) Normal $\rightarrow$ Poisson.

(13) Normal $\rightarrow$ Binomial.

(14) Exponential as limit of Geometric.